🔑 **Answer Key — Teacher Copy · Lesson 8-5**

MSTAR PRACTICE WORKSHEET

Lesson 8-5: Understand Inequalities and Their Solutions

Grade 6 Mathematics · Standard 6.AT.9

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: A

Which graph represents $x < 5$?

- **A** Open circle at 5, shaded left
- **B** Closed circle at 5, shaded left
- **C** Open circle at 5, shaded right
- **D** Closed circle at 5, shaded right

> $x < 5$ means 5 is NOT included (open circle) and values less than 5 are shaded (left).

Question 2 SELECTED RESPONSE — CORRECT: A

Which graph represents $x \geq 3$?

- **A** Closed circle at 3, shaded right
- **B** Open circle at 3, shaded right
- **C** Closed circle at 3, shaded left
- **D** Open circle at 3, shaded left

> $x \geq 3$ means 3 IS included (closed circle) and values greater than or equal to 3 are shaded (right).

Question 3 SELECTED RESPONSE — CORRECT: A

Which graph represents $x > 8$?

- (A) Open circle at 8, shaded right
- (B) Closed circle at 8, shaded right
- (C) Open circle at 8, shaded left
- (D) Closed circle at 8, shaded left

> $x > 8$ means 8 is NOT included (open circle) and values greater than 8 are shaded to the right.

Question 4 SELECTED RESPONSE — CORRECT: A

Which graph represents $x \le 10$?

- (A) Closed circle at 10, shaded left
- (B) Open circle at 10, shaded left
- (C) Closed circle at 10, shaded right
- (D) Open circle at 10, shaded right

> $x \le 10$ means 10 IS included (closed circle) and values less than or equal to 10 are shaded (left).

Question 5 SELECTED RESPONSE — CORRECT: A

A number line shows a closed circle at 8 and is shaded to the left. Which inequality does this represent?

- (A) $x \le 8$
- (B) $x < 8$
- (C) $x \ge 8$
- (D) $x > 8$

> Closed circle means the value is included ($\le$ or $\ge$). Shaded left means less than or equal to: $x \le 8$.

Question 6 SELECTED RESPONSE — CORRECT: A

Clue: the witness ran more than 15 feet, so $d > 15$. On the number line, what do you draw?

- **A** Open circle at 15, shade right
- **B** Closed circle at 15, shade right
- **C** Open circle at 15, shade left
- **D** Closed circle at 15, shade left

> 'Greater than' (>) is a strict symbol, so 15 is NOT included — open circle. Larger values work, so shade right.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: A

A number line shows an open circle at 7, and the shading covers the numbers to the right of 7. Which inequality does this graph represent?

- **A** $x > 7$
- **B** $x \geq 7$
- **C** $x < 7$
- **D** $x \leq 7$

> Shading to the right means the solution set holds the numbers greater than 7. The open circle means the boundary point 7 is left out, so the inequality is $x > 7$.

- **B:** A closed circle would include 7. The circle on this graph is open.
- **C:** That shades to the left. The shading here covers the numbers to the right of 7.
- **D:** That includes 7 and shades left. The circle is open and the shading runs right.

Part B — correct answer: C

Why is the circle at 7 open instead of closed?

- **A** An open circle is used whenever the shading points to the right.
- **B** An open circle shows that 7 is included in the solution set.
- **C** The symbol > does not include 7, so 7 is not in the solution set and the boundary point stays unfilled.
- **D** Every inequality graph uses an open circle at its boundary point.

> An open circle marks a boundary point that is not a solution. Substituting the boundary gives $7 > 7$, which is false, so 7 is excluded and the circle is left unfilled. A closed circle would be used for ≥, where the boundary is a solution.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, D

Select ALL numbers that belong to the solution set of $y \leq 12$.

- A 12
- B 11.5
- C 12.5
- D 0
- E 20

> The symbol ≤ includes the boundary point, so 12 is a solution and its circle would be filled in. Numbers below 12, such as 11.5 and 0, are solutions as well. The values 12.5 and 20 are greater than 12, so they are outside the solution set.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A theme park has two rides. Ride A requires height $h > 48$ inches. Ride B requires height $h \geq 48$ inches. A child is exactly 48 inches tall. Which ride can they go on? Explain using number line graphs for both inequalities.

> **Model answer:** The child is exactly on the boundary, so test 48 in each rule. For Ride A, $48 > 48$ is false, so the graph has an open circle at 48 and the child cannot ride. For Ride B, $48 \geq 48$ is true, so the graph has a closed circle at 48 and the child can ride B.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.
